

CHAPTER 4

AI GOVERNANCE BY HUMAN RIGHTS-CENTERED DESIGN, DELIBERATION, AND OVERSIGHT

An End to Ethics Washing

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Finding ways of developing and deploying new technologies with a purpose restricted to supporting individual freedom and dignity as well as the basic constitutional settlement of constitutional democracies, namely democracy, rule of law and fundamental rights is the challenge of our time.

—Paul Nemitz (2018)

INTRODUCTION

The number and variety of topics in this volume illustrate the width, diversity of content, and vagueness of the boundaries of AI (artificial intelligence) ethics as a domain of inquiry.¹ Within this discourse, increasing attention has been drawn to the capacity of

¹ The opening quote is from Paul Nemitz, principal adviser, European Commission's Directorate-General for Justice and Consumers, "Profiling the European Citizen: Why Today's Democracy Needs to Look Harder at the Negative Potential of New Technology Than at Its Positive Potential," in *Being*

socio-technical systems that utilize data-driven algorithms to classify, to make decisions, and to control complex systems, including the use of machine learning and large datasets to generate predictions about future behavior (hereafter “AI” systems)², may interfere with human rights. The recent Cambridge Analytica scandal revealed how unlawfully harvested Facebook data from millions of voters in the United Kingdom, the United States, and elsewhere enabled malign actors to engage in political micro-targeting through the use of AI-driven social media content distribution systems, thereby interfering with their right to free and fair elections and thus threatening the integrity of democratic processes. The increasing use of algorithmic decision-making (ADM) systems to inform custodial and other decisions within the criminal justice process may threaten several human rights, including the right to a fair trial, the presumption of innocence, and the right to liberty and security. Systems of this kind are now used to inform, and often to automate, decisions about an individual’s eligibility and entitlement to various benefits and opportunities, including housing, social security, finance, employment and other life-affecting opportunities, potentially interfering with rights of due process and rights to freedom from unfair or unlawful discrimination.³ Because these systems have the capacity to operate both automatically and at scale, their capacity to affect thousands if not millions of people at a stroke can now occur at orders of magnitude and speeds not previously possible.⁴

This chapter has two overarching aims. Firstly, we argue that the international human rights framework provides the most promising set of standards for ensuring that AI systems are ethical in their design, development, and deployment. Secondly, we sketch the basic contours of a comprehensive governance framework, which we refer to as a human rights-centered design, deliberation, and oversight approach for ensuring that AI can be relied upon to operate in ways that will not violate human rights.

Four features of ongoing discussions provide important contexts for our argument. First, the rubric of “AI ethics” is now used to encapsulate a multiplicity of value-based, societal concerns associated with the use of AI applications across an increasingly extensive and diverse range of social and economic activities. Second, there is a notable

Profiled: Cogitas Ergo Sum, ed. Bayamlioglu, Irina Baraliuc, Liisa Janssens, and Mireille Hildebrandt (Amsterdam: Amsterdam University Press, 2018).

² For more detail, see European Commission, “Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions on Artificial Intelligence for Europe,” 2018, <https://ec.europa.eu/digital-single-market/en/news/communication-artificial-intelligence-europe> (accessed March 17, 2020).

³ B. Wagner, *Study on the Human Rights Dimensions of Automated Data Processing Techniques (in Particular Algorithms) and Possible Regulatory Implications*, Council of Europe, Committee of experts on internet intermediaries (MSI-NET), 2017, <https://rm.coe.int/study-hr-dimension-of-automated-data-processing-incl-algorithms/168075b94a> (accessed June 11, 2018).

⁴ Karen Yeung, *A Study of the Implications of Advanced Digital Technologies (Including AI Systems) for the Concept of Responsibility within a Human Rights Framework*, Council of Europe MSI-AUT committee study, 2019, DGI(2019)05, <https://rm.coe.int/a-study-of-the-implications-of-advanced-digital-technologies-including/168096bdab> (accessed December 9, 2019).

lack of clarity about the content of the normative values and principles that constitute the relevant “ethical” standards to which AI systems should adhere. Third, industry self-regulation is the predominant approach for bringing about “ethical AI,” reflected in a litany of “ethical codes of conduct” promulgated by individual tech firms and various tech industry consortia published in response to the recent “Tech Lash.”⁵ These codes presuppose that the tech industry can formulate appropriate ethical norms for AI and can be trusted to ensure that AI systems will duly adhere to those standards. Suggestions to apply more conventional regulation, involving legally mandated regulatory standards and enforcement mechanisms, are swiftly met by protests from the industry that “regulation stifles innovation.”⁶ These protests assume that innovation is an unvarnished and unmitigated good, based on an unexamined belief that technological innovation (particularly in the digital services industry) should be relentlessly pursued without regard to its adverse impacts. This belief is now entrenched as the altar upon which cash-strapped contemporary governments worship, naïvely hoping that digital innovation will create jobs, stimulate economic growth, and thereby fill diminishing governmental coffers left bare after propping up the banking sector that teetered on the brink of collapse following the global financial crisis in 2008. Fourth, discussion of the need for meaningful *enforcement* of ethical standards is almost entirely absent from these initiatives.⁷

This chapter proceeds in three stages. First, we argue that international human rights standards offer the most promising basis for developing a coherent and universally recognized set of standards that can be applied to meet many (albeit not all) of the normative concerns currently falling under the rubric of AI ethics. Second, the paper outlines the core elements of a human rights-centered design, deliberation, and oversight approach to the governance of AI, explaining why such an approach is needed. Because much more theoretical and applied research is required to flesh out the details of our proposed approach, the third section sets out an agenda for further research, identifying the multiple lines of inquiry that must be pursued to develop the technical and organizational methods and systems that will be needed, based on the adaptation of existing engineering and regulatory techniques aimed at ensuring safe system design, reconfiguring and extending these approaches to secure compliance with a much wider and more complex set of human rights norms. The fourth and final section concludes with reflections on the limitations of our proposed approach and the magnitude of the challenges associated with making it implementable in real world settings. Nevertheless, we suggest that a human rights-centered design, deliberation, and oversight approach to

⁵ Eve Smith, “The Techlash against Amazon, Facebook and Google—and What They Can Do,” *The Economist* (Jan. 20, 2018), <https://www.economist.com/briefing/2018/01/20/the-techlash-against-amazon-facebook-and-google-and-what-they-can-do> (accessed June 11, 2018).

⁶ There have, however, been more recent concessions by Big Tech about the need for legal regulation: e.g., Microsoft’s call for legal regulation of facial recognition technology by states, and Facebook’s Mark Zuckerberg’s recent acknowledgment that some kind of regulation is needed in relation to data-driven social media content distribution systems.

⁷ An exception is recent legislation and proposed legislation (e.g., in Germany, France and the United Kingdom) concerned with reducing the prevalence of extremist and terrorist media content online.

the governance of AI offers a concrete proposal capable of delivering *genuinely* ethical AI, for at least four reasons, to which we now turn.

WHY SHOULD HUMAN RIGHTS LIE AT THE CORE OF AI ETHICS?

Within contemporary discussions of AI ethics, there is no agreed set of ethical standards that should govern the operation of AI systems, reflected in the variety of ethical standards espoused in various voluntary AI ethics codes that have emerged in recent years. The salience of AI ethics reflects welcome recognition by the tech industry and policymakers that AI systems may have significant adverse impacts,⁸ with some values commonly appearing in these discussions, particularly those of “transparency,” “fairness,” and “explainability.”⁹ Yet the vagueness and elasticity of the scope and content of AI ethics has meant that it currently operates as an empty vessel into which anyone (including the tech industry and the so-called Digital Titans) can pour their preferred ethical content. Without an agreed framework of norms that clearly identifies and articulates the relevant ethical standards that AI systems should be expected to comply with, little real progress will be made toward ensuring that these systems are in practice designed, developed, and deployed in ways that will meet widely accepted ethical standards. Although there is scope for reasonable disagreement concerning what ethical conduct requires in any given case, a core set of agreed norms that constitute the basic minimum below which conduct cannot fall if it can be appropriately characterized as ethically acceptable must be identified.¹⁰

We believe that international human rights standards offer the most promising set of ethical standards for AI systems, as several civil society organizations have suggested.¹¹

⁸ Yeung, *supra* n. 4.

⁹ See Daniel Greene, Anna Lauren Hoffmann, and Luke Stark, “Better, Nicer, Clearer, Fairer: A Critical Assessment of the Movement for Ethical Artificial Intelligence and Machine Learning,” in *Proceedings of the 52nd Hawaii International Conference on System Sciences*, <https://dblp.org/rec/bibtex1/conf/hicss/GreeneHS19> (accessed December 11, 2019); Luciano Floridi, Josh Cows, Monica Beltrametti, Raja Chatila, Patrice Chazerand, Virginia Dignum, and Christoph Luetge et al., “AI4People—An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations,” *Minds and Machines* 28 (2018): 689–707.

¹⁰ P. Nemitz, “Constitutional Democracy and Technology in the Age of Artificial Intelligence,” *Phil Trans. A.* 376, no. 2133 (2018): 20180089.

¹¹ See various reports by civil society organizations concerned with securing the protection of international human rights norms.; e.g., M. Latonero, “Governing Artificial Intelligence: Upholding Human Rights and Human Dignity,” *Data & Society* (2018), https://datasociety.net/wp-content/uploads/2018/10/DataSociety_Governing_Artificial_Intelligence_Upholding_Human_Rights.pdf (accessed May 6, 2019); *The Toronto Declaration: Protecting the Rights to Equality and Non-Discrimination in Machine Learning Systems* (2018), <https://www.accessnow.org/the-toronto-declaration-protecting-the-rights-to-equality-and-non-discrimination-in-machine-learning-systems/>; *The Montreal Declaration for a Responsible Development of Artificial Intelligence: A Participatory Process*

As an international governance framework, human rights law is intended to establish global standards (norms) and mechanisms of accountability that specify the ways in which individuals are entitled to be treated. The United Nations (UN) Universal Declaration of Human Rights (UDHR) of 1948 is perhaps the most well-known international human rights charter, based on a commitment that the appalling treatment of individuals that occurred during World War II should not only be condemned and prohibited outright but ought never be repeated. Despite the number of, and variation between, regional human rights instruments in the Americas, Africa, and Europe, and enshrined in the constitutions of individual nation-states, they are all grounded on a shared commitment to uphold the inherent human dignity of each and every person in which each individual is regarded of equal dignity and worth, wherever situated.¹² These shared foundations reflect the status of human rights standards as *basic moral entitlements* of every individual by virtue of their humanity, whether or not those entitlements are given explicit legal protection.¹³

The extent to which governments recognize these basic moral entitlements as *legally enforceable* rights varies considerably, partly due to differences in political ideology. In contemporary liberal democratic states, human rights are now widely recognized as essentially “constitutional” in status to provide effective guarantees that individual freedoms will be cherished and respected. In particular, the European Union’s (EU) legal order is rooted in constitutional commitments to human rights, democracy, and the rule of law, the so-called constitutional triumvirate that forms the foundational principles upon which political systems characterized as liberal constitutional democracies ultimately rest.¹⁴ This brings us to a second reason why human rights norms provide the appropriate norms for securing ethical AI: because a commitment to effective human rights protection is part and parcel of democratic constitutional orders. In a world in which AI systems increasingly configure our collective and individual environments, entitlements, and access to, or exclusion from, opportunities and resources, it is essential that the protection of human rights, alongside respect for the rule of law and the protection of democracy, is assured to maintain the character of our political communities as constitutional democratic orders in which every individual is free to pursue his or her own version of the good life as far as this is possible within a framework of peaceful and stable cooperation underpinned by the rule of law.¹⁵ This contrasts starkly with

(2017), <https://nouvelles.umontreal.ca/en/article/2017/11/03/montreal-declaration-for-a-responsible-development-of-artificial-intelligence/>; *Access Now* (see <https://www.accessnow.org/tag/artificial-intelligence/> for various reports); *Data & Society* (see <https://datasociety.net/>); *IEEE Report* (see <https://www.ieee.org/>) on ethically aligned design for AI (<https://ethicsinaction.ieee.org/>), which lists as its first principle that AI design should not infringe international human rights; *AI Now Report* (2018), https://ainowinstitute.org/AI_Now_2018_Report.pdf. See also L. McGregor et al., “International Human Rights Law as a Framework for Algorithmic Accountability,” 68 *ICLQ* (2019): 309–43.

¹² Latonero, *supra* n. 11.

¹³ R. Dworkin, *Taking Rights Seriously* (London: Duckworth, 1977).

¹⁴ P. Nemitz, *supra* n. 10 at 376.

¹⁵ M. Hildebrandt, *Smart Technologies and the End(s) of Law: Novel Entanglements of Law and Technology* (Cheltenham, UK: Edward Elgar, 2015).

most contemporary AI ethics codes, which typically outline a series of “ethical” principles that have been effectively plucked out of the air, without any grounding in a specific vision of the character and kind of political community that its authors are committed to establishing and maintaining and that those principles are intended to secure and protect.¹⁶

The well-developed institutional framework through which systematic attempts are made to monitor, promote, and protect adherence to human rights norms around the world provide two additional reasons in support of adopting human rights standards to ensure the ethical governance of AI systems. Despite considerable variation in the range and scope of rights enumerated in formal Charters of Rights, there is a well-established analytical framework through which tension and conflict between rights, and between rights and collective interests of considerable importance in democratic societies, are resolved in specific cases through the application of a structured form of reasoned evaluation. This approach is exemplified in the structure and articulation of human rights norms within the European Convention on Human Rights (ECHR). The ECHR (ratified by forty-seven countries) specifies a series of human rights norms, including (among others) the right to freedom of expression, the right to life, the right to private and home life, and the right to freedom of assembly and religion, all of which must be guaranteed to all individuals and effectively protected. However, for many of those rights, certain qualifications are permitted in order to ensure respect for a narrow range of clearly specified purposes that are necessary in a democratic society, provided that any such qualifications are prescribed by law and proportionate in relation to those purposes. So, for example, Article 10 of the ECHR provides that:

- (1) Everyone has the right to freedom of expression. This right shall include freedom to hold opinions and to receive and impart information and ideas without interference by public authority and regardless of frontiers. This Article shall not prevent States from requiring the licensing of broadcasting, television or cinema enterprises.
- (2) The exercise of these freedoms, since it carries with it duties and responsibilities, may be subject to such formalities, conditions, restrictions or penalties as are prescribed by law and are necessary in a democratic society, in the interests of national security, territorial integrity or public safety, for the prevention of disorder or crime, for the protection of health or morals, for the protection of the reputation or rights of others, for preventing the disclosure of information received in confidence, or for maintaining the authority and impartiality of the judiciary.

Accordingly, although freedom of expression is essential to ensure, among other things, free democratic debate and individual self-determination, legal restrictions on expression may be permissible for the purposes specified in Article 10(2). Accordingly, restrictions

¹⁶ See for example the *Beijing AI Principles*, <https://www.baai.ac.cn/blog/beijing-ai-principles> (accessed December 6, 2019).

on expression may be justified in order, for example, to protect individual rights to privacy or the right to free and fair elections, if they conflict in particular cases, provided that restrictions are legally prescribed and go no further than the minimum necessary to protect these rights.

This structured framework for reasoned resolution of conflict arising between competing rights and collective interests in specific cases is widely understood by human rights lawyers and practitioners, forming an essential part of a human rights approach. This framework overcomes another shortcoming in existing codes of ethical conduct: their failure to acknowledge potential conflicts between ethical norms, and the lack of any guidance concerning how those conflicts will or ought to be resolved in the design and operation of AI systems. Of the codes that do acknowledge potential conflict, little is offered by way of guidance concerning how to resolve such conflict: both in the codes themselves, and in much of the ongoing AI ethics literature, beyond suggesting one should seek help from an ethics expert.¹⁷

In contrast, the well-established human rights approach to the resolution of ethical conflict is informed by, and developed through, a substantial body of authoritative rulings handed down by judicial institutions (at both international and national levels) responsible for adjudicating human rights complaints. These adjudicatory bodies, which determine allegations of human rights violations lodged by individual complainants, form part of a larger institutional framework that has developed over time to monitor, promote, and protect human rights, and includes a diverse network of actors in the UN system, other regional human rights organizations (such as the Council of Europe and a wide range of civil society organizations focused on the protection of human rights), national courts and administrative agencies, academics, and other human rights advocates. The institutional framework for rights monitoring, oversight, and adjudication provides a further reason why human rights norms provide the most promising basis for AI ethics standards. The dynamic and evolving corpus of judicial decisions can help elucidate the scope of justified interferences with particular rights in concrete cases, offering concrete guidance to those involved in the design, development, and implementation of AI systems concerning what human rights compliance requires. Most importantly, perhaps, these human rights norms are both internationally recognized and, in many jurisdictions, supported by law, thereby providing a set of national and international institutions through which allegations of human rights violations can be investigated and enforced, and hence offer a means for real and effective protection.

This contrasts sharply with the prevailing self-regulatory model favored by the tech industry and to which most national and regional governments (including the EU) have

¹⁷ See Council of Europe, “Guidelines on Artificial Intelligence and Data Protection,” January 2019, <https://rm.coe.int/guidelines-on-artificial-intelligence-and-data-protection/168091f9d8>. While the EU High Level Expert Group’s *Ethics Guidelines for Trustworthy AI* (2019) refers to the need for reasoned evaluation, transparency, and documentation of how such ethical trade-offs are resolved when they are encountered by those involved in designing, developing, and deploying AI systems, it offers no substantive guidance concerning how that evaluation should be conducted.

acquiesced.¹⁸ Although self-regulation has been effective in a handful of industries, which can be understood as a “community of shared fate” (e.g., the U.S. nuclear industry after Three Mile Island via the Institute of Nuclear Power Operations (INPO)),¹⁹ there are good reasons to doubt their effectiveness in general,²⁰ given that self-regulatory standards that have no legally binding force.²¹ Because tech firms operate in highly competitive global markets in which securing the first-mover advantage is often accompanied by the capacity to reap the extensive benefits arising from global network effects (e.g., Google Maps), it is naïve to expect that they can be trusted to abide by voluntary standards when faced with such powerful commercial imperatives. It is hardly surprising that critics have dismissed these voluntary codes of conduct as “ethics washing”²² given overwhelming evidence that the tech industry cannot be relied upon to honor its voluntary commitments.²³ Nemitz describes the growth of these initiatives as a “genius move” by the tech industry, allowing the industry to focus attention and resources on the ethics of AI to delay the debate and work on the *law* for AI.²⁴ As Hagendorff comments:

AI ethics—or ethics in general—lacks mechanisms to reinforce its own normative claims. Of course, the enforcement of ethical principles may involve reputational

¹⁸ Self-regulation is the controlling of a process or activity by the people or organizations that are involved in it rather than by an outside organization such as government. Voluntary self-regulation or “pure” self-regulation entails the private firm or industry making and enforcing the rules, independent of direct government involvement: N. Gunningham, “Investigation of Industry Self-Regulation in Workplace Health and Safety in New Zealand,” 2011, http://regnet.anu.edu.au/sites/default/files/publications/attachments/2015-04/NG_investigation-industry-self-regulation-whss-nz_o.pdf (accessed June 14, 2019).

¹⁹ J. Rees, *Hostages to Each Other: The Transformation of Nuclear Safety since Three Mile Island* (Chicago: University of Chicago Press, 1994).

²⁰ For a list of shortcomings, see Organisation for Economic Co-operation and Development (OECD), *Industry Self-Regulation: Role and Use in Supporting Consumer Interests*, DSTI/CP(2014)4/FINAL 20–21 (Paris: OECD). Regulation expert Neil Gunningham observes that “The extent to which self-regulation in practice has either positive or negative attributes will depend very much on the social and economic context within which it operates and on the particular characteristics of the scheme itself. Nevertheless it is fair to say that ‘pure’ self-regulation is rarely effective in achieving social objectives because of the gap between private industry interests and the public interest.” See Gunningham, *supra* n 18, 3.

²¹ F.Z. Borgesius, *Discrimination, Artificial Intelligence and Algorithmic Decision-Making*, Council of Europe, Directorate General for Democracy (Strasbourg: Council of Europe, 2018), <https://rm.coe.int/discrimination-artificial-intelligence-and-algorithmic-decision-making/1680925d73> (accessed June 3, 2019).

²² B. Wagner, “Ethics as an Escape from Regulation: From Ethics-Washing to Ethics-Shopping?” in Bayamlioglu et al. *supra* n 1.

²³ For a sobering account of Facebook’s repeated failure to honor its publicly stated commitments, see UK House of Commons, Digital Culture Media and Sports Committee, *Disinformation and “Fake News”: Final Report*, Eighth Report of Session 2017–2019, February 14, 2019, HC 1791, <https://publications.parliament.uk/pa/cm201719/cmselect/cmcumeds/1791/1791.pdf> (accessed May 6, 2019).

²⁴ Nemitz, *supra*, n. 10, 376.

losses... Yet... these mechanisms are rather weak and pose no eminent [*sic*] threat... Ethical guidelines of the AI industry serve to suggest to legislators that internal self-governance is sufficient, and that no specific laws are necessary to mitigate possible technological risks and to eliminate scenarios of abuse... And even when more concrete laws concerning AI systems are demanded, as recently done by Google... these demands remain relatively vague and superficial.²⁵

Though a handful of ethical AI proposals advocated by civil society and other international organizations have drawn attention to the need to ensure that AI systems respect human rights norms, they have paid scant attention to their enforcement.²⁶

WHY A HUMAN RIGHTS-CENTERED DESIGN, DELIBERATION, AND OVERSIGHT OF AI?

The ineffectiveness of the prevailing self-regulatory approach to “ethical AI” demonstrates that an alternative governance model is needed: (1) one that is anchored in human rights norms and a human rights approach, (2) one that utilizes a coherent and integrated suite of technical, organizational, and evaluation tools and techniques, (3) one that is subject to legally mandated external oversight by an independent regulator with appropriate investigatory and enforcement powers, and (4) one that provides opportunities for meaningful stakeholder and public consultation and deliberation. In the next section, we develop a governance framework for AI systems intended to do just that, which we call “human rights-centered design, deliberation and oversight.” Although

²⁵ T. Hagendorf, “The Ethics of AI Ethics: An Evaluation of Guidelines” (2019), <https://arxiv.org/abs/1903.03425> (accessed May 6, 2019). As the recent EU “Algo-Aware” (Dec. 2018) project observes, “Across the globe, the majority of initiatives (ie concerned with programmes aimed at securing algorithmic accountability) are very recent or still in development. Additionally, there are limited concrete legislative or regulatory initiatives being implemented.” Algo-Aware, *State of the Art Report: Algorithmic Decision-Making*. (2018), <https://actuary.eu/wp-content/uploads/2019/02/AlgoAware-State-of-the-Art-Report.pdf> (accessed June 7, 2019).

²⁶ For example, *The Toronto Declaration: Protecting the Rights to Equality and Non-Discrimination in Machine Learning Systems*, which focuses only on rights to equality and non-discrimination in ML systems, draws attention to the character of human rights as a “universally ascribed system of values based on the rule of law” that constitute a “universally binding, actionable set of standards” (per paragraph 9) for which “prompt and effective remedies” must be available against “those responsible for violations” (per paragraph 49): See <https://www.accessnow.org/the-toronto-declaration-protecting-the-rights-to-equality-and-non-discrimination-in-machine-learning-systems/>. Although the EU’s *Ethical Guidelines for Trustworthy AI* places human rights protection at their foundation, offering an “Assessment List” in order to provide concrete guidance to tech firms seeking to adhere to the ethical guidance thereby provided, the guidelines remain entirely voluntary, and make no provision for external, independent oversight and enforcement.

much more foundational work remains to be done, both to specify the content and contours of this approach more fully and to render it capable of practical implementation, we believe that our proposed framework offers a concrete approach that can bring an end to “ethics washing” by securing the design, development, and deployment of human rights-compliant AI systems in real-world settings. The core elements of our approach are outlined in the following discussion.

What Is Human Rights–Centered Design, Deliberation, and Oversight?

Our proposed governance regime for AI and other relevant automated systems (understood as complex sociotechnical systems, which includes the data upon which they rely for training and operation) seeks to ensure that these systems will be human rights-compliant and reflect the core values that underpin the rule of law.²⁷ It entails systematic consideration of human rights concerns at every stage of system design, development, and implementation (making interventions where this is identified as necessary). Such a regime should be mandated by law; should be subject to external oversight by independent, properly resourced regulatory authorities with appropriate powers of investigation and enforcement; and should provide for input from both technical and human rights experts, on the one hand, and meaningful input and deliberation from affected stakeholders and the general public on the other. Our approach seeks to integrate ethical design strategies, technical tools and techniques for software and system design, verification, testing, and auditing together with social and organizational approaches to effective and legitimate governance. In so doing, our approach seeks to integrate a range of methods from a wide variety of intersecting disciplinary perspectives, including the following perspectives.

1. Pragmatic “ethics in design” frameworks have been developed by applied ethicists concerned with ensuring that due attention is given to moral values in technical innovation processes early on in the technical design process, with the aim of integrating values

²⁷ Our proposed approach to AI governance can be understood as compatible with, and complementary to, Hildebrandt’s concept of “legal protection by design (LPbD).” Hildebrandt’s LPbD places greater emphasis on articulating the challenges for legal protection and the rule of law posed by code-driven technologies (including an identification of the substantive and procedural opportunities and capacities which computational systems must provide in order to ensure that the values and commitments underpinning the rule of law are reflected in these systems, such as rights of contestation, rights to demand an explanation and justification, etc.). In contrast, our ‘human rights-centred design, deliberation, and oversight’ approach places greater emphasis on developing concrete legal, technical, and organizational governance methods and techniques for ensuring that human rights protection is implemented into complex socio-technical systems that utilize AI technologies: See Hildebrandt, *supra* n. 15; M. Hildebrandt, *Law for Computer Scientists* (Oxford: Oxford University Press, 2019), <https://lawforcomputerscientists.pubpub.org> (accessed June 19, 2019).

and engineering design.²⁸ For example, the Value-Sensitive Design (VSD) approach developed by Friedman and others builds on the insights of the human-computer interaction community, with a concern to incorporate human and moral values into the design of information technology, connecting those who design systems with those affected by them and other stakeholders.²⁹

2. Another perspective involves engineering techniques concerned with “hard-wiring” specific values into a sociotechnical system’s design and operation. Although the most established methods and experience have hitherto focused on ensuring the value of safety, primarily via safety engineering techniques, an increasing body of work in software design and engineering has expanded the range of values that engineers have sought to encode and protect via system design. These include privacy (referred to as “privacy-enhancing technologies” or “privacy by design”), security (“security by design”), and, more recently, data protection principles (“data protection by design”). In the realm of machine learning, a growing body of technical research in “explainable AI” (XAI) has been devoted to enhancing the capacity for machine learning systems to provide avenues through which humans can better understand the logic by which machine learning systems generate outputs as well as techniques for improving the “fairness, accountability and transparency” (FAT) of these outputs by seeking to identify and eliminate unfair discrimination. Taken together, these techniques can be understood as falling within this expanding family of technical approaches to securing ethical values beyond that of safety.

3. In an additional approach, a suite of methods and techniques in software design and engineering, including various forms of assessment, testing, and evaluation, can identify whether particular aspects of a system (provably) meet certain prespecified standards and requirements.³⁰

4. Organizational accountability mechanisms and established regulatory techniques used in safety critical domains operate *ex ante*, requiring systematic evaluation of safety concerns and appropriate interventions *before* a system is deployed. They also involve *ex post* techniques that apply after the system has been deployed and that have been designed and developed to ensure the traceability and auditability of system behavior via systematic recording and logging of a system’s design and operation and any alterations thereto.³¹

5. A range of regulatory governance techniques have been used effectively in other contexts, including the use of (i) impact assessment tools and methods that incorporate

²⁸ J. van den Hoven, S. Miller, and T. Pogge (eds.), *Designing in Ethics* (Cambridge: Cambridge University Press, 2017), 28.

²⁹ B. Friedman, “Value-Sensitive Design” *Interactions* 3 (1996): 16–23; B. Friedman, P. Kahn, and A. Borning, *Value-Sensitive Design: Theory and Methods*, CSE Technical Report 02-12-01 (Seattle: University of Washington, 2002).

³⁰ J. A. Kroll et al., “Accountable Algorithms,” *U. Pa. L. Rev.* 165 (2017): 633.

³¹ A. Rieke, M. Bogen, and D. G. Robinson, *Public Scrutiny of Automated Decisions: Early Lessons and Emerging Methods*, An Upturn and Omidyar Network Report (2018), <https://www.omidyar.com/insights/public-scrutiny-automated-decisions-early-lessons-and-emerging-methods> (accessed June 3, 2019).

opportunities and mechanisms for facilitating stakeholder consultation, engagement, and deliberation, particularly in relation to the design, development, and deployment of high-risk applications;³² (ii) risk-based approaches to regulation that seek to ensure that high-risk systems are subject to the most intensive and demanding scrutiny, whilst the burdens of demonstrating human rights compliance for low-risk systems are proportionately less demanding;³³ (iii) meta-regulatory approaches that seek to harness the knowledge and expertise within firms themselves in the service of regulatory compliance, overseen by a public regulator endowed with powers of investigation and sanction; and (iv) post-implementation monitoring (“AI system vigilance”), in order to systematically and transparently track adverse events in order to identify problems and failures as early as possible and facilitate swift corrective interventions.

Our expectation is that these frameworks and methods can be adapted and refined to ensure that respect for *human rights norms* is integrated into system design, while incorporating a human rights approach to the resolution of conflict and tension between human rights norms, or between human rights norms and important collective interests, which may arise in specific contexts and circumstances. At the same time, we anticipate the need for new techniques and frameworks to accommodate novel human rights risks that the development and deployment of AI systems may generate. For example, these are likely to include new governance frameworks and oversight mechanisms to ensure that data-driven experimentation on human users when undertaken outside conventional academic research settings is undertaken in a human rights-compliant manner, while the necessity of “in the wild testing” of AI systems generate novel governance challenges that do not arise in circumstances where the product and service development phase can be sharply delineated from their deployment.

Core Principles of Human Rights–Centered Design, Deliberation, and Oversight

The methods and techniques listed in the preceding section vary widely in their disciplinary foundations and in the original contexts of their development. Our proposal seeks to draw them together in an integrated manner, appropriately adapted toward ensuring conformity with human rights norms, as the basis for a comprehensive design and governance regime constructed around the following four core principles in which human rights norms provide the foundational ethical standards that AI systems must demonstrably comply with:

³² See, for example, data protection impact assessment, A. Mantelero, “AI and Big Data: A Blueprint for a Human Rights, Social and Ethical Impact Assessment,” *Computer Law & Security Review* 34, no. 4 (2018): 754–772; and human rights impact assessment, F. A. Raso et al., *Artificial Intelligence & Human Rights: Opportunities & Risks* (Berkman Klein Center, 2018).

³³ J. Black, *Risk-Based Regulation: Choices, Practices and Lessons Being Learned: Risk and Regulatory Policy* (Paris: OECD, 2008).

1. Design and deliberation
2. Assessment, testing, and evaluation
3. Independent oversight, investigation, and sanction
4. Traceability, evidence, and proof

Each of these principles is briefly outlined in the following discussion.

Principle 1: Design and deliberation

Central to our approach is a requirement that AI systems should be designed and configured to operate in ways that are compliant with universal human rights standards (such as those, for example, set out in the ECHR). At least for systems identified during the design and development phase as posing a “high risk” of interfering with human rights, affected stakeholders should be consulted about the proposal and given opportunities to express their views about the proposed system’s potential impact, in discussion with the system’s designers. Consultation with affected stakeholders and the general public during the initial phases of system design contributes to the overall legitimacy of the regime, understood in terms of respect for democratic values and affected communities, and should help system designers to identify which aspects of the system’s design and proposed operation need reconsideration. Where the risks to human rights are assessed as “high” or “very high,”³⁴ this would trigger an obligation on system designers to reconsider and redesign the system and/or proposed business model³⁵ in order to reduce those risks to a form and level regarded as tolerable (understood in terms of a human rights approach to the resolution of conflict between rights and collective interests), in ways that duly accommodate concerns expressed by affected stakeholders and in recognition of the individual and collective benefits that the system is expected to generate.³⁶

Principle 2: Assessment, testing, and evaluation

Users and others affected by the operation of AI systems (including the general public) can only have justified confidence that AI systems do in fact comply with human rights standards if these systems can be subjected to formal assessment and testing to evaluate their compliance with human rights standards, and if these occur regularly throughout the entire lifecycle of system development: from the initial formulation of a proposal through to design, specification, development, prototyping, and real-world

³⁴ S. Jasanoff, *The Ethics of Invention: Technology and the Human Future* (New York: W. W. Norton, 2018); Raso et al., *supra* n. 32.

³⁵ On the potential discriminatory impact of data-driven business models, see M. Ali et al., “Discrimination through Optimization: How Facebook’s Ad Delivery Can Lead to Skewed Outcomes” (2019), <https://arxiv.org/pdf/1904.02095.pdf> (accessed June 3, 2019).

³⁶ The participatory approach to social impact assessment referred to in the Council of Europe’s AI Guideline strongly resonates with the role that our approach ascribes to public deliberation: see The Council of Europe, Guidelines on AI (Feb. 19, 2019), <https://rm.coe.int/guidelines-on-artificial-intelligence-and-data-protection/168091f9d8> at 23–24.

implementation and including periodic evaluation of the datasets upon which the system has been trained and upon which it operates.³⁷

These evaluations form a core element of an overarching “human rights risk-management” approach, which aims to identify potential human rights risks *before* the deployment of AI and other relevant automated systems and that occurs within a larger meta-regulatory approach to AI governance in which AI system developers and owners are subject to legal duties to demonstrate to a public regulatory authority that their system is human rights-compliant.³⁸ If significant risks to human rights compliance are identified, system developers must reconsider the design specification and system requirements with a view to modifying them in order to reduce those risks to a level that satisfies the tests of necessity and proportion³⁹—or, in cases where the threats to human rights are disproportionate and thus unacceptably high, to refrain from proceeding with the development of the system in the form proposed. Once the system has been implemented, periodic review must be undertaken and test and assessment documents duly filed with the public authority. A system of “AI vigilance” is also needed, entailing the systematic recording of adverse incidents arising from system operations, including potential human rights violations reported by users or the wider public, triggering an obligation on the system provider to review and reassess the system’s design and operation, and to report and publicly register any modifications to the system undertaken following this evaluation. Systematic and periodic post-implementation monitoring and vigilance is needed to ensure that AI systems continue to operate in a human rights-compliant manner because, once implemented into real-world settings, AI systems will invariably display emergent effects that are both difficult to anticipate and may scale very rapidly. Accordingly, there is also an accompanying need for more systematic and sustained research concerned with modeling social systems in order to better anticipate and predict their unintended adverse societal effects.

³⁷ Kroll et al., *supra* n. 30; Borgesius, *supra* n. 21; Rieke, Bogen, and Robinson, *supra* n. 31.

³⁸ Also called “management-based” regulation and “enforced self-regulation,” meta-regulation refers to a strategy in which regulators do not prescribe how regulated firms should comply, but instead require them to develop their own systems for compliance with legally mandated goals and to demonstrate that compliance to the regulator: J. Black, “Paradoxes and Failures: ‘New Governance’ Techniques and the Financial Crisis,” *Modern Law Review* 75 (2012): 1037, 1045–1048.

³⁹ The formulation of the appropriate legal standard would need to reflect the established proportionality assessment that is well-established in addressing human rights conflicts and conflicts between human rights and legitimate collective interests, operating as the human rights equivalent as the “as low as reasonably practicable” (ALARP) requirement that applies to legal duties to ensure the safety of complex systems, per A. Hopkins, “Explaining Safety Case,” Regulatory Institutions Network Working Paper 87 (2012), https://www.csb.gov/assets/1/7/workingpaper_87.pdf; M. Thomas, “Safety-Critical Systems,” Gresham Lectures (London, 2017), <https://www.gresham.ac.uk/lectures-and-events/safety-critical-systems> (accessed March 18, 2020).

Principle 3: Independent oversight, investigation, and sanction

In order to provide meaningful assurance that AI systems are in fact human rights-compliant, rather than merely *claiming* to be human rights-compliant, independent oversight by an external, properly resourced, technically competent oversight body invested with legal powers of investigation and sanction is essential.⁴⁰ Because the operation of market forces cannot provide those who design, develop, and deploy AI systems with sufficient incentives to invest the required resources necessary to ensure that AI systems are human rights-compliant, our proposed approach must operate within a *legally mandated* institutional structure, including an oversight body with a duty to monitor and enforce substantive and procedural (regulatory) requirements, including those concerning robust design, verification, testing, and evaluation (including appropriate documentation demonstrating that these requirements have been fulfilled), supported by legally mandated stakeholder and public consultation where proposed AI systems pose a “high risk” to human rights.

We suggest that independent oversight is best designed within a meta-regulatory framework, in which legal duties are placed on AI system developers and operators to demonstrate to a public authority that their systems are human rights-compliant.⁴¹ Although there are a variety of approaches that can be understood as meta-regulatory in form,⁴² the so-called safety case, properly implemented, is considered to have significantly contributed to ensuring the safety of complex systems in several domains, including safety regulation for offshore petroleum drilling through to the regulation of workplace safety adopted in several Anglo-Commonwealth legal systems.⁴³

⁴⁰ Borgesius, *supra* n. 21.

⁴¹ See n. 39 above.

⁴² See J. Black, “Managing Regulatory Risks and Defining the Parameters of Blame: A Focus on the Australian Prudential Regulation Authority,” *Law and Policy* 28 (2006): 1–26; S. Gilad, “It Runs in the Family: Meta-Regulation and Its Siblings,” *Regulation & Governance* 4 (2010): 485–506; C. Coglianese and E. Mendelson, “Meta-Regulation and Self-Regulation,” in R. Baldwin, C. Hood, and M. Lodge (eds.), *The Oxford Handbook of Regulation* (New York: Oxford University Press, 2010): 146–168.

⁴³ The so-called “safety case” movement emerged in the early 1990s in both the United Kingdom and the United States as an approach to safety certification involving approval and oversight of complex systems, such as aircraft, nuclear power plants, and offshore oil exploration. See Hopkins, *supra* n. 39. There have, however, been criticisms of a safety case approach, including concerns about problems of confirmation bias, the need to consider worst case scenarios, reliance on probabilistic assessment to provide assurances of safety rather than the opposite goal of identifying unrecognized hazards, and examples of highly successful process-based (rather than performance-based) approaches to securing safety in relation to submarines (e.g., the SUBSAFE program): see N. Leveson, “The Use of Safety Cases in Certification and Regulation,” MIT Engineering Systems Division Working Paper Series (2011), <http://sunnyday.mit.edu/SafetyCases.pdf> (accessed June 12, 2019). Leveson observes that the British Health and Safety Executive has applied a safety case regime widely to UK industries, pursuant to which responsibility for controlling risks is placed primarily on those who create and manage hazardous systems, based on three principles: (a) those who create the risks are responsible for controlling those risks, (b) safe operations are achieved by setting and achieving goals rather than by following prescriptive rules, (c) while those goals are set out in legislation, it is for the system providers and operators to develop what they consider to be appropriate methods to achieve those goals.

In his discussion of offshore petroleum drilling, Hopkins highlights five basic features of a safety case approach:

- (1) All operators must prepare a systematic risk (or hazard) management framework, which identifies all major hazards and provides detailed plans for how these hazards will be managed, specifying the controls that will be put in place to deal with the identified hazards and the measures that will be taken to ensure that controls continue to function as intended.
- (2) A requirement for the operator to “make the case” to the regulator, that is, to demonstrate to the regulator that the processes that have been undertaken to identify hazards, the methodology they have used to assess risks, and the reasoning (and evidence) that has led them to choose one control rather than another, should be regarded as acceptable. It is then up to the regulator to accept (or reject) the case. Although a safety case gives operators considerable independence and flexibility in determining how they will respond to hazards, they do not have free rein: thus if an operator proposes to adopt an inadequate standard, a safety case regulator may challenge the operator and require the adoption of a better standard.
- (3) A competent, independent, and properly resourced regulator with the requisite level of expertise and who can engage in meaningful scrutiny. The regulator’s role is not to ensure that hardware is working, or that documents are up to date, but to *audit against the safety case*, to ensure that the specified controls are functioning as intended, and this necessitates a sophisticated understanding of accident causation and prevention.
- (4) Employee participation, both in the development of safety cases, and with whom the regulatory officials carrying out site audits must consult.
- (5) A general legal duty of care imposed on the operator to do whatever is reasonably practicable to identify and control all hazards. An operator cannot claim to be in compliance just because it has completed a hazard identification process. It is the general duty of care that raises a safety case regime above a “tick box” or “blind compliance” mentality, so that a hazard identification process that is demonstrably inadequate would fail to meet the requisite standard.⁴⁴

Regulatory regimes of this kind allow (although they do not necessitate) the possibility of ex ante licensing by a designated public authority in the case of particularly human rights-sensitive, high-risk systems, such as facial recognition systems intended for use by governments to identify individuals of interest in public places.⁴⁵ Applying the

⁴⁴ Although the general duty of care is linguistically quite imprecise, its meaning has been elaborated on via case law, through numerous cases in which courts have had to decide whether the duty has been complied with. This case law gives fairly clear guidance as to what the general duty means in particular cases: see Hopkins *ibid.*, *supra*, n. 39.

⁴⁵ See for example Big Brother Watch, *Face Off: The Lawless Growth of Facial Recognition in UK Policing* (2018), <https://bigbrotherwatch.org.uk/wp-content/uploads/2018/05/Face-Off-final-digital-1.pdf> (accessed June 12, 2019).

underlying logic and structure of the safety case approach to human rights compliance would provide developers with considerable flexibility in seeking to “make the case” to the regulator to demonstrate that their proposed AI systems can be expected to operate in human rights-compliant ways.

Principle 4: Traceability, evidence, and proof

In order to facilitate meaningful independent oversight and evaluation, AI systems must be designed and built to secure auditability: this means more than merely securing transparency, but is aimed at ensuring that they can be subject to *meaningful review*, thus providing a concrete evidential trail for securing human accountability over AI systems.⁴⁶ Not only is it necessary that systems be constructed to produce evidence that they operate as desired,⁴⁷ there must be a legal obligation to do so, requiring that crucial design decisions, the testing/assessment process and the outcome of those processes, and the operation of the system itself are properly documented and provide a clear evidence trail that can be audited by external experts. Drawing again on the experience of the safety case approach, which entails imposing a legal duty on operators to demonstrate to the regulator that robust and comprehensive systems are in place that reduce safety risks to a level that is “as low as reasonably practical,” we envisage the imposition of a suitably formulated legal duty on AI systems developers, owners, and operators to demonstrate that these systems are human rights-compliant.

To discharge this legal duty, AI system developers would also be subject to legal duties to prepare, maintain, and securely store system design documentation, testing, and evaluation reports, and the system must be designed to routinely generate operational logs that can be inspected and audited by an independent, suitably competent authority. Taken together, these provide an audit trail through which system designers and developers can demonstrate that they have undertaken human rights “due diligence”—thereby discharging their legal duty to demonstrate that they have discharged their legal duty to reduce the risk of human rights violations to an acceptable level. These traceability and evidential requirements apply to both the design and development phase (including verification and validation requirements) and the operation and implementation of systems (logging and black-box recording of system operations). Taken together, these obligations are intended to ensure that robust and systematic transparency mechanisms are put in place, the aim of which is not complete comprehension but to provide sufficient information to ensure that human accountability for AI systems can be maintained.⁴⁸ This integrated approach to AI governance grounded on these

⁴⁶ J. J. Bryson and A. Theodorou, “How Society Can Maintain Human-Centric Artificial Intelligence,” in *Human-Centered Digitalization and Services*, ed. M. Toivonen-Noro and E. Saari (Singapore: Springer Nature Singapore, 2019), 12–13.

⁴⁷ Kroll et al., *supra* n. 30.

⁴⁸ Bryson and Theodorou, *supra* n. 46 at 14.

principles can be understood as a response to the Council of Europe's call for a "human-rights oriented development of technology."⁴⁹ As Alessandro Mantalero has claimed:

Innovation must be developed responsibly, taking the safeguard of fundamental rights as the pre-eminent goal... This necessarily requires the development of assessment procedures, the adoption of participatory models and supervisory authorities. A human rights-oriented development of tech might increase costs and force developers and business to slow their current time-to-market, as the impact of products and services on individual rights and society have to be assessed in advance. At the same time, in the medium to long-term, this approach will reduce costs and increase efficiency (e.g., more accurate prediction and decision systems, increased trust, fewer complaints). Moreover, businesses and societies are mature enough to view responsibility towards individuals and society as the primary goal in AI development.⁵⁰

GETTING FROM HERE TO THERE: A RESEARCH AGENDA

The four principles outlined in the previous section demand revision to many aspects of software engineering (SE) practice. While a suite of relevant engineering and regulatory governance techniques are already in use in some specific areas, they require significant adaptation and generalization to support meaningful human rights evaluation and compliance. Changes to SE practice must be complemented by a focused human rights-centered design research agenda in computer science. Such an agenda would draw together the currently fragmented activity in relevant software engineering disciplines (including SE, cybersecurity, HCI, verification) and also consider their continued relevance to the software lifecycle of AI systems in particular, which are likely to require new design processes. Rather than offer a detailed research agenda here, we offer instead a "manifesto," which identifies and briefly outlines some of the technical, engineering, and governance challenges that must be met if SE is to provide assurances of human rights compliance. Some of these topics are existing areas of practice in software engineering and others are established research disciplines. None, however, have any tradition of human rights-centered design and most are only just beginning to consider appropriate software engineering practice for AI systems.

Requirements Analysis

In SE, requirements analysis concerns the identification of the needs to be met by a new software system. The commercial orientation and diversity of approaches to requirements

⁴⁹ A. Mantalero, *AI and Data Protection: Challenges and Possible Remedies*, Study for Council of Europe (2018), <https://rm.coe.int/artificial-intelligence-and-data-protection-challenges-and-possible-re/168091f8a6> (accessed June 3, 2019).

⁵⁰ See Mantalero *ibid.* section 1.3 for more detail.

analysis present a severe challenge to Principle 1 of our human rights-centered design agenda, and it is made more acute by the fact that requirements analysis has been developed for non-AI systems. Corporate practice is strongly oriented to identifying the requirements of business customers and are often contractual in nature: accordingly, requirements specification often generates long lists of “shoulds” stated in a natural language. In contrast, where consumers are targeted end-users, their work, play, and social needs are typically the focus of requirements analysis. Smaller tech companies, particularly start-ups, identify requirements analysis as the major component of SE concerns⁵¹ and are heavily dependent on agile- (or even craft-) based approaches to development that have weaker contractual requirements analysis and weaker audit trails. However, it has at least been realized that affected stakeholders, *beyond* those of the customer and end-user, should be identified and involved as participants in requirements analysis and design, giving rise to participatory design.⁵² It is also the case that professionals involved in requirements analysis have a diversity of backgrounds—not only computer science and engineering but also psychology, sociology, and other social sciences, and therefore might be extended to include those with legal training. In order to meet Principle 1, software engineering practice of AI systems must meet the following challenges:

- a. How to consider human rights requirements for all stakeholders, not only as users or customers but also as individual rights-bearers entitled to equal concern and respect, in auditable requirements description and requirements specification documents?
- b. How to train and employ design professionals who can bring human rights-centered design methods to system design and requirements specification?

Understanding, Collecting, and Analyzing Data

The processes for acquiring, selecting, and modeling data that are required by AI systems create their own human rights challenges. These challenges require attention at several levels, including the way in which problems are framed during requirements analysis. A human rights-centered approach to design that meets Principle 1 must take due account of human rights risks when building AI systems requirements. For example, many commercial AI systems are designed to utilize data-driven “hypernudges” to channel user attention and action in directions beneficial to the system owner.⁵³ These potentially threaten individual autonomy, dignity, and the right to liberty and freedom of thought, yet these human rights risks are not currently taken into account in requirements analysis processes. *Bias in data sampling, modeling, and attribute selection is*

⁵¹ E. U. Klotins and T. Gorschek, “Software Engineering Antipatterns in Start-Ups,” *IEEE Software* 36, no. 2 (2019): 118–126.

⁵² J. Simonsen and T. Robertson ed., *Routledge International Handbook of Participatory Design* (New York: Routledge, 2013).

⁵³ K. Yeung, “‘Hypernudge’: Big Data as a Mode of Regulation by Design,” *Information, Communication & Society* 20, no. 1 (2017): 118–136.

another major problem. The use of machine learning techniques generates many opportunities for bias and discrimination to inadvertently affect the outputs they produced, which may threaten the right to equal protection (the right to nondiscrimination).⁵⁴ These include biases of the algorithms' developers, biases built into the models upon which the systems are generated, biases inherent in the datasets used to train the models, or biases introduced when such systems are implemented in real-world settings.⁵⁵ In response to these concerns, a growing body of work concerned with devising technical approaches for countering such biases has emerged, but this has yet to move out of the lab into software development settings.

Verification

Verification concerns processes for checking whether the software meets specified requirements. In other words, does the software satisfy the output of the requirements analysis? Verification can involve the use of formal methods (logic) to check that software, or a model of software, does not contain errors. Formally verified software performs the required functions and nothing else for all possible inputs with verifiable evidence. Verification is used, for example, in the aviation industry and to some extent in other safety-critical systems. Verification has also seen some successes in more agile software development environments.⁵⁶ It does not guarantee that, for example, a plane will not crash but can guard against undesirable conditions occurring by virtue of misconceived models or poorly written software. The application of verification is mandated by certification authorities in some sectors (e.g., aviation) but not in others. It is also used in some sectors (e.g., ship design) because the commercial costs of errors are relatively high. Where it is mandated, processes are typically subject to audit by a government authority, such as the Federal Aviation Administration (FAA) or the European Aviation Safety Agency (EASA). Verification is particularly relevant to Principles 1 (design and deliberation) and 2 (assessment, testing, and evaluation). We ask two questions: (1) Can AI systems be formally verified? (2) Can verification be human rights-centered? The answer to the first question is negative, at least with current methods,⁵⁷ although

⁵⁴ Protocol No 12 ECHR Article 1 provides that "the enjoyment of any right set forth by law shall be secured without discrimination on any ground such as sex, race, color, language, religion, political or other opinion, national or social origin, association with a national minority, property, birth or other status." See also Art 21 EU Charter of Fundamental Rights.

⁵⁵ M. Veale and R. Binns, "Fairer Machine Learning in the Real World: Mitigating Discrimination without Collecting Sensitive Data," *Big Data & Society* (Dec. 2017), <https://doi.org/10.1177/2053951717743530>.

⁵⁶ C. Calcagno, D. Distefano, J. Dubreil, D. Gabi, P. Hooimeijer, M. Luca, P. O'Hearn, I. Papakonstantinou, J. Purbrick, and D. Rodriguez, "Moving Fast with Software Verification," in *NASA Formal Methods: 7th International Symposium*, ed. K. Havelund, G. Holzmann, and R. Joshi, Lecture Notes in Computer Science 9058 (Cham, Switzerland: Springer, 2015), 3–11.

⁵⁷ S. Russell, D. Dewey, and M. Tegmark, "Research Priorities for Robust and Beneficial Artificial Intelligence," *AI Magazine* 36, no. 4 (2015): 105–114.

this is an active area of research. Note that AI is not used in safety-critical systems precisely because the software cannot be verified. AI systems are difficult to verify for several reasons. In particular, programs automatically generated by machine learning are coded in different forms to hand-coded computer programs, and existing verification methods are not designed to work with such programs. Relatedly, these programs are typically more complex and have a much higher level of dimensionality in comparison with hand-coded programs, so verification approaches may not be computationally tractable. In addition, machine learning can be used in deployed systems to adapt their behavior in real time, so that effective verification would need to be continually repeated in the use context. It is therefore impossible for current verification techniques to be human rights-centered. Nevertheless, human rights-centered verification may still play a role in the design of validation procedures (more on this in the following section). Human rights-centered verification of AI systems is likely to require many years of research before it influences practice. This research should be designed to answer the following challenges:

- a. What are the limits of verification with respect to AI systems and requirements concerning human rights beyond safety?
- b. How can formal methods be used to verify an AI systems substrate, such as the operating systems and learning software,⁵⁸ at least to ensure that AI systems are operating as intended?
- c. How can formal methods be used to advance AI systems testing (more on this follows)?

Cybersecurity by Design

Cybersecurity concerns the protection of computer systems, data, and cyber-physical systems from intrusion, theft, or damage. Cybersecurity by design focuses on the need for security from software foundations and therefore for security considerations in the requirements analysis. Further, some have advocated the use of, for example, formal verification methods early in the design process.⁵⁹ Unfortunately, cybersecurity by design has not been a consideration in software engineering until relatively recently and many deployed systems and practices suffer as a consequence. However, this is changing, often in response to legislation, and early efforts at building a regulatory and engineering infrastructure may provide a way forward for human rights-centered design. Inevitably cybersecurity by design for AI systems faces the same challenges as verification discussed earlier. Further challenges are documented in a recent NSF report, although that report focuses on privacy rather than on human rights in general.

⁵⁸ Ibid.

⁵⁹ S. Chong, J. Guttman, A. Datta, A. Myers, B. Pierce, P. Schaumont, T. Sherwood, and N. Zeldovich, *Report on the NSF Workshop on Formal Methods for Security*, arXiv preprint arXiv:1608.00678 (2016).

Validation

Validation methods are used to assess whether the behavior of a software system meets stakeholders' needs. Validation is not simply about checking the behavior of the system against the written specification; it is very much a rigorous empirical process that must generate data relevant to understanding whether a system is fit for a purpose. Two specific forms of validation that are conducted extensively in the software industry are:

(a) Penetration testing

Penetration testing is a commissioned cyberattack, conducted by an internal or external agency. It is one method used to ensure the security of software systems and data. Penetration testing is sometimes automated and there are standard tests that are legally mandated in some industries (e.g., the payment card industry). A particular focus for penetration testing is privacy validation. However, the value of privacy is sometimes regarded as an absolute value, to be protected at all costs, which may not reflect the rights-balancing approach enshrined in how the right to privacy is understood within a human rights approach. Penetration testing has not, to our knowledge, been applied to AI systems but recent demonstrations of how AI systems can be spoofed with adversarial attacks⁶⁰ suggest that these systems will come with new, unanticipated vulnerabilities. In order to meet the needs of Principle 2, penetration testing must meet the following challenges:

- (i) Acknowledge and explicitly address trade-offs. Current practice and literature emphasizes privacy in ways that may disproportionately threaten the protection of other rights or legitimate collective interests when they come into conflict in specific contexts and circumstances.
- (ii) Address the problem of how to counter the threat of Adversarial AI (also known as Offensive AI), particularly as it is likely to be applied to AI systems.⁶¹

(b) User Experience Design

User experience (UX) designers often play key roles in requirements analysis and also in empirical validation. Tasks and systems (artifacts) often co-evolve, and UX designers provide critical feedback on the effectiveness of existing designs as well as ideas for future designs. Typically, they focus on ensuring that system use is useful, pleasurable, rewarding, and efficient. UX designers are also tasked with seeing things from the user's perspective rather than from the service provider's perspective. Methods include the use of scenarios and personas that provide means to curate stories about context of use and potential users. Participatory design has grown in importance, providing one way in which human values from outside the industry can influence design. Human-centered design research has had a strong influence on UX design practice, but methods are

⁶⁰ M. Hutson, "Hackers Easily Fool Artificial Intelligences," *Science* (July 20, 2018): 215.

⁶¹ M. Brundage, et al. *The Malicious Use of AI: Forecasting, Prevention, and Mitigation* (2018), <https://maliciousaireport.com/> (accessed December 11, 2019).

not currently configured to embrace human rights concerns in the form recognized under international human rights law. Much has been made of the need for UX designers to consider the social and physical context of use,⁶² and there are a number of ethical and value-motivated influences,⁶³ including bias,⁶⁴ feminism,⁶⁵ accessibility, and diversity research. But, little work is done on the democratic context of use and therefore on human rights from a legal and constitutional perspective. Accordingly, key questions include: What is the future of UX design for AI systems? How can UX design move beyond its current focus on social and physical contexts to embrace democratic and civic structures (including respect for human rights) as important sources of constraint? Can methods used in the human rights community for engaging people with human rights thinking⁶⁶ contribute to UX approaches to AI systems design?

Appropriation

Software systems are not only designed, they are also appropriated by users for unanticipated tasks and unanticipated contexts. This productive aspect of human use of technology may prove particularly problematic for the governance of AI systems. ADM systems have hitherto been built for one specific social context, but this context is rarely communicated in a robust way that assures that it is only used in this setting. For example, as Zweig and Krafft discuss, the software COMPAS is used in criminal justice systems for pre-trial assessment (such as bail decisions or decisions to prosecute) but was originally built for post-trial assessment.⁶⁷ Further work is needed to investigate how appropriation of AI systems can be appropriately governed (Principle 3).

Algorithmic Transparency and Inspection

At present there is no general requirement for algorithm inspection, but recent cases suggest that a systematic approach should be considered in order to provide assurance of the trustworthiness of algorithmic systems, particularly those that directly and

⁶² C. Heath and P. Luff, *Technology in Action* (Cambridge: Cambridge University Press, 2000); D. Benyon, P. Turner, and S. Turner, *Designing Interactive Systems: People, Activities, Contexts, Technologies* (Harlow, UK: (Harlow: Pearson Education, 2005).

⁶³ B. Friedman and D. G. Hendry, *Value Sensitive Design: Shaping Technology with Moral Imagination* (Cambridge, MA: MIT Press, 2019); B. Friedman ed., *Human Values and the Design of Computer Technology*, (Cambridge: Cambridge University Press, 1997).

⁶⁴ B. Friedman and H. Nissenbaum, "Software Agents and User Autonomy," in *Agents '97: First International Conference on Autonomous Agents* (ACM, 1997), 466–469.

⁶⁵ S. Bardzell, "Feminist HCI: Taking Stock and Outlining an Agenda for Design," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (ACM, 2014), 1301–1310.

⁶⁶ See European Union External Action, "Good Human Rights Stories Coalition Launched" (Sept. 28, 2018), https://eeas.europa.eu/headquarters/headquarters-homepage/51241/good-human-rights-stories-coalition-launched_en (accessed December 11, 2019).

⁶⁷ K. A. Zweig, G. Wenzelburger, and T. D. Krafft, "On Chances and Risks of Security Related Algorithmic Decision-Making Systems," *European Journal for Security Research* 3 (2018): 181–203.

adversely affect the rights of individuals. In the context of ADM systems in the U.S. criminal justice system, this has been especially problematic where the algorithms have been developed by commercial software providers claiming intellectual property protection, enabling them to assert rights of confidentiality and secrecy over their algorithms.⁶⁸ In contrast, the development and implementation of the HART algorithm used in the United Kingdom by the Durham police force to make custody decisions has been much more open; it has not, however, been made available for public scrutiny.⁶⁹ While there are legitimate concerns about “gaming” that may justify refraining from full public disclosure of certain algorithms, at least in high-risk contexts where human rights are seriously threatened, regulators must have legal powers to inspect algorithms and datasets (supporting Principle 3).⁷⁰

Instrumentation and Logging

Bryson and Theodorou argue that logging should be mandated in all “socially critical” fields.⁷¹ We assert that logging must also be mandated in all “human rights critical” fields (i.e., all systems identified that pose a “high risk” of unjustifiably interfering with human rights, particularly when they can do so at scale). Firms are unlikely to keep an audit trail that evidences their problematic actions unless they are legally required to do so (e.g., mandatory black-box recorder requirements in the aviation industry). The importance of maintaining audit trails of system behavior is essential for maintaining meaningful human accountability over AI systems: human rights-centered design demands instrumentation in systems so that they automatically record and reproduce historical decision-making processes and outcomes: we should mandate this, at minimum, for all safety-critical and human rights-critical systems (in support of Principle 4).

CONCLUSION

This chapter has highlighted various deficiencies inherent in the prevailing model of voluntary self-regulation for securing “ethical AI.” It has enabled a “Pick Your Own”

⁶⁸ See *State of Wisconsin v. Loomis* (2016) 881 N.W.2d 749 (Supreme Court of Wisconsin).

⁶⁹ Durham Police have been reported that they “would be prepared to reveal the HART algorithm and the associated personal data and custody event datasets to an algorithmic regulator”: M. Burgess, “UK Police Are Using AI to Inform Custodial Decisions—But It Could Be Discriminating against the Poor,” *Wired* (March 1, 2018), <https://www.wired.co.uk/article/police-ai-uk-durham-hart-checkpoint-algorithm-edit> (accessed December 11, 2019).

⁷⁰ But see the Canadian Directive on Automated Decision-Making 2019, which applies to the Canadian federal government’s use of automated decision-making systems including machine learning and predictive analytics which, among other things, imposes requirements for the release of any custom source code that is owned by the Government of Canada.

⁷¹ Bryson and Theodorou, *supra* n. 46.